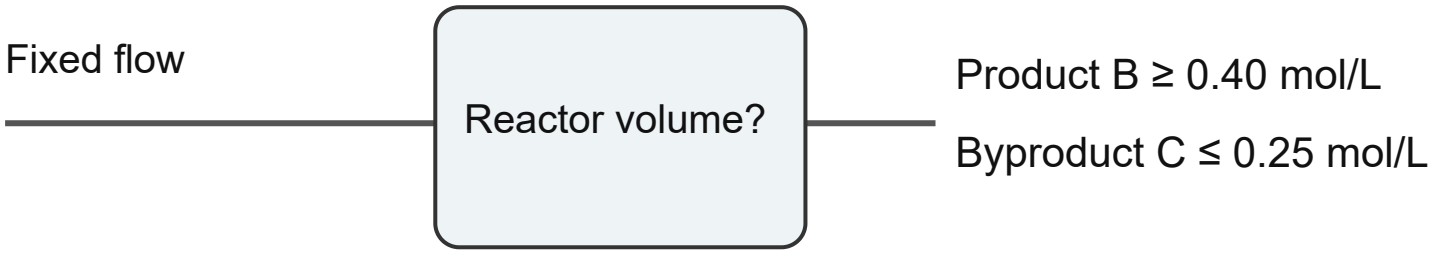
Week 5: From a Predictor to an Optimization Formulation

The smallest reactor that meets product specifications

Product B must reach 0.40 mol/L while byproduct C stays below 0.25 mol/L. At fixed flow, how small can the reactor be?



Changing residence time changes both B and C.

Product specifications and reactor volume minimization

Given a residence time, the network predicts outlet concentrations. Design asks for a residence time that satisfies those concentration requirements.

Longer residence converts A to B, but also converts B to C. Maximizing B and minimizing volume are different objectives.

This week: decision formulation and frozen ReLU computation using linear constraints and binary variables.

Learning objectives for optimization formulation

1. Decisions, fixed context, predicted outputs, objective, and constraints
2. Big-M equivalence for one ReLU: proof by branch
3. Valid pre-activation bounds from the input domain
4. Hand-built MILP versus forward pass at identical inputs

Submission: formulation, verification results, and solution interpretation.

Decision variables and fixed context

Role	Value or range in the lab
Decision	Residence time $\tau \in [2, 10]$ min
Fixed context	$T = 330$ K, $C_{Af} = 1.2$ mol/L
Fixed flow	$Q = 1$ L/min
Predicted output	A, B, C concentrations y [mol/L]
Objective	Minimize volume $V = Q\tau$ [L]

Here τ is a design variable selecting volume at fixed flow. Choosing temperature as well requires a model with temperature as an input and bounds over that domain.

Choosing residence time fixes the required volume through $V=Q\tau$ at fixed flow. Changing residence time in an existing vessel instead requires changing flow or another condition, which changes the variables and costs of the design problem.

An optimization problem in symbols

$$\min_{u,y} J(u, y; d)$$
$$y = f_\theta(u; d),\ g(u, y; d) \le 0$$

Notation: decisions u, fixed context d, model outputs y. Fitted weights θ: fixed.

Feasible point: satisfies network equations and process constraints. Optimum: feasible point with the smallest objective.

Prediction supplies u and computes y. Optimization treats both as unknowns and links them with y=fθ(u). Encoding that relation prevents the solver from changing y independently just to satisfy specifications.

CSTR design optimization problem

$$\min_{\tau, y} V = Q\tau$$
$$2 \leq \tau \leq 10, y = f_\theta(\tau)$$
$$y_B \geq .40, y_C \leq .25, y \geq 0$$

With Q = 1 L/min, minimizing τ also minimizes V. Outputs are in mol/L and must be converted to those units before checking specifications.

B ≥ 0.40 is a concentration specification. Purity or production-rate requirements need constraints written from their own definitions.

τ=2 gives a small volume but may lack sufficient B. Increasing τ until B reaches its lower bound identifies a boundary below which volume is smaller but B fails. The C upper limit must also hold at that boundary before it is a minimum feasible design.

Reference concentrations for the lab

Assume perfect mixing, constant volume, equal inlet and outlet flow, and consecutive 1:1 reactions A → B → C. At 330 K, $k_1 = 0.2\ \text{min}^{-1}$ and $k_2 = 0.1\ \text{min}^{-1}$.

$$C_A = \frac{1.2}{1 + .2\tau}$$

$$C_B = \frac{.2\tau C_A}{1 + .1\tau}, C_C = .1\tau C_B$$

We evaluate these equations at five residence times to construct a small ReLU model.

The frozen computational graph

$$h^0 = x$$
$$a^\ell = W^\ell h^{\ell-1} + b^\ell$$
$$h^\ell = \max\left(0, a^\ell\right)$$
$$y = W_{\text{out}} h^K + b_{\text{out}}$$

ReLU: componentwise in each hidden layer. Optimization variables: inputs, intermediate values, and outputs. Numeric constants: W and b.

Making weights variable introduces products of W and h, so the MILP derived here would no longer apply.

Intermediate network values become optimization variables, but they are not free choices. Each affine equality and ReLU constraint ties one layer to the preceding layer. This converts the forward calculation into relations that must hold simultaneously.

Equality representation of input scaling

$$x = \frac{\tau - 2}{8}, 0 \le x \le 1$$
$$\tau = 2 + 8x$$

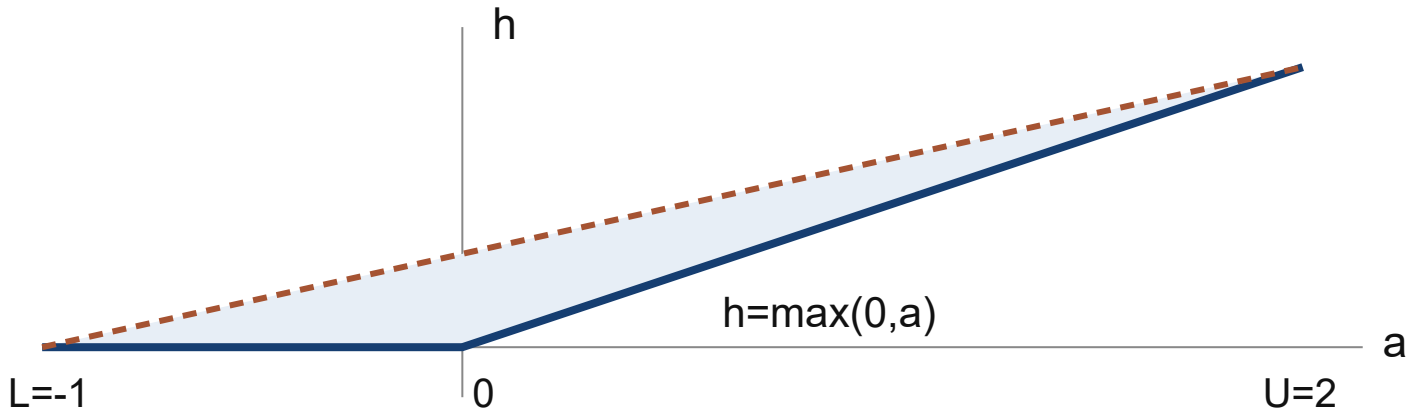
The lab maps τ to dimensionless input x. Optimization uses exactly the offset and scale used to construct the predictor.

If outputs were standardized as well, include y = μ + Ds. This inverse transformation is linear when μ and D are fixed.

For example, τ=6 gives x=(6−2)/8=0.5. Feeding 6 directly to the network leaves its 0–1 input domain. Check the coordinates entering each equation rather than relying only on variable names.

Piecewise-linear structure of ReLU

$$h = \max(0, a)$$



The blue line is the exact graph. The shaded region includes points admitted when the binary variable is relaxed.

For a<0, h=0 is horizontal; for a>0, h=a is a line of slope one. Binary z selects the branch. Relaxing z to a continuous value also admits points between the branches, allowing outputs above the actual ReLU.

The ReLU graph and its lower inequalities

$$h \geq 0, h \geq a$$

At a = −0.2, h = 0.4 satisfies both inequalities, although ReLU returns 0.

The objective or a later layer may favor larger h. For an arbitrary network, these lower inequalities alone are insufficient; exact graph modeling also needs branch-specific upper inequalities.

Big-M formulation of one ReLU

Assume valid finite bounds $L \le a \le U$ with $L < 0 < U$.

$$h \ge 0, h \ge a$$
$$h \le a - L(1 - z)$$
$$h \le Uz, z \in \{0, 1\}$$

Branches: inactive at z = 0; active at z = 1. L and U: bounds on this neuron's pre-activation.

$h \le Uz$ pins h to zero when z=0. $h \le a - L(1-z)$ pins h to a when z=1. Valid L and U make the unused upper inequality permissive enough to retain every correct point on the selected branch.

Equivalence of the inactive branch (z = 0)

$$h \leq U0 = 0, h \geq 0$$
$$h = 0, a \leq h = 0$$

The remaining upper inequality, $0 \leq a - L$, holds because $a \geq L$. This branch therefore represents exactly $L \leq a \leq 0$ and $h = 0$.

Equivalence of the active branch (z = 1)

$$h \le a - L0 = a, h \ge a$$
$$h = a \ge 0$$

The remaining upper inequality h ≤ U is equivalent to a ≤ U. Thus 0 ≤ a ≤ U and h = a.

At a = 0, z can be either 0 or 1, and the output h is fixed at 0.

Substituting numbers into one ReLU formulation

For $a=2x-1$ and $x\in[0,1]$, $L=-1, U=1$. The inequalities are $h\geq 0$, $h\geq a$, $h\leq a+1-z$, and $h\leq z$.

Input and branch	Lower bound	Upper bound	Output
$x=0.2$, $a=-0.6$, $z=0$	$h\geq 0$	$h\leq 0$; $h\leq 0.4$	$h=0$
$x=0.8$, $a=0.6$, $z=1$	$h\geq 0.6$	$h\leq 0.6$; $h\leq 1$	$h=0.6$
Wrong $z=0$ at $x=0.8$	$h\geq 0.6$	$h\leq 0$	Impossible

```
a = 2 * x - 1
h = max(0., a)
z = int(a >= 0.)
assert h >= 0 and h >= a
assert h <= a + 1 - z and h <= z
```

The binary selects a branch compatible with a; it does not create an arbitrary output. At $a=0$, either z is possible but both give $h=0$.

Linear representations of stable active and inactive neurons

Valid range	Replacement	Binary variable
$U \le 0$	$h = 0$	None
$L \ge 0$	$h = a$	None
$L < 0 < U$	Four Big-M inequalities	One

Allowing inputs outside the bounded domain invalidates this simplification. Recheck bounds and classifications when changing the domain.

An interval $[-2,-0.1]$ never activates, so replace the neuron by $h=0$. On $[0.2,1.3]$, $h=a$ everywhere. Only a possible sign change needs branch selection; bounds therefore determine which neurons require binaries.

Interval bounds accounting for weight signs

$$a = \sum_i w_i h_i + b$$
$$L = b + \sum_i \min\left(w_i l_i, w_i u_i\right)$$
$$U = b + \sum_i \max\left(w_i l_i, w_i u_i\right)$$

Assume $l_i \le h_i \le u_i$ for every input. Negative weights reverse which input endpoint gives the minimum or maximum.

Interval arithmetic gives valid bounds, but ignoring correlations between inputs can make them wider than the attainable range.

A bound calculation by hand

$$a = 2x_1 - 3x_2 + .5$$
$$0 \le x_1 \le 1, -1 \le x_2 \le 2$$
$$L = -5.5, U = 5.5$$

The minimum occurs at $(x_1, x_2) = (0, 2)$, and the maximum at $(1, -1)$. Do this calculation before choosing an arbitrary large constant.

For the lower bound, use $x_1=0$ with positive coefficient 2 and $x_2=2$ with negative coefficient −3: $0-6+0.5=-5.5$. Reverse those endpoints for the upper bound: $2-3\times(-1)+0.5=5.5$. A negative coefficient reverses the direction of an extreme.

Bound validity and formulation correctness

Using $U = 0.5$ at a point whose true pre-activation is $a = 0.7$ excludes $h = 0.7$: even $z = 1$ imposes $h \leq 0.5$.

Wider valid bounds preserve the graph at integer solutions, but can weaken the continuous relaxation.

Observed extrema in the training samples do not automatically bound the entire input domain.

Frozen coefficients of the ReLU interpolant

Linearly interpolate reference concentrations at τ = 2, 4, 6, 8, 10 min. Fix the hidden hinges at those breakpoints and compute output weights from interpolation slopes.

$$a_j = x - t_j, h_j = \max\left(0, a_j\right)$$

$$t = \left[\begin{array}{cccc} 0 & .25 & .50 & .75 \end{array}\right]$$

$$y_k = b_k + \sum_{j=0}^{3} W_{\mathrm{jk}} h_j$$

j = 0, 1, 2, 3 and output k is A, B, or C. This lab constructs the model by interpolation rather than gradient training.

Apply the first segment slope from the start. At each later knot, an activating ReLU adds the difference between the new and old slopes. The active slope changes therefore sum to the slope of the current segment.

Neuron bounds in the lab model

Neuron	a_j	L_j	U_j	Treatment
0	x	0	1	$h_0 = x$
1	$x - 0.25$	-0.25	0.75	Binary
2	$x - 0.50$	-0.50	0.50	Binary
3	$x - 0.75$	-0.75	0.25	Binary

There are three binary variables. Substituting the affine pre-activations leaves eight continuous variables: one x, four h values, and three outputs.

Hidden values and B concentration at τ=5 min

Subtract thresholds [0,0.25,0.50,0.75] from x=(5−2)/8=0.375.

j	$a_j=x-t_j$	h_j	Contribution $W_jB\cdot h_j$
0	0.375	0.375	0.380952381×0.375=0.142857143
1	0.125	0.125	−0.268398268×0.125=−0.033549784
2	−0.125	0	0
3	−0.375	0	0

Adding bias 0.285714286 gives B≈0.395021645 mol/L. Taking the midpoint of the 4-min and 6-min values also gives (0.380952381+0.409090909)/2=0.395021645.

The ReLU sum and segment interpolation express the same calculation. They differ from reference-process B=0.4 at 5 min, so this prediction falls slightly below the B≥0.40 specification.

Matrix representation of Big-M constraints

For neuron 1, $a_1 = x - 0.25$, $L_1 = -0.25$, and $U_1 = 0.75$.

$$h_1 - x \geq -.25$$
$$h_1 - x + .25z_1 \leq 0$$
$$h_1 - .75z_1 \leq 0$$

Set $h_1 \geq 0$ and $0 \leq z_1 \leq 1$ as variable bounds, and declare z_1 integer. The lab assembles the full matrix in the same way.

The second row collects $h_1 \leq x - 0.25 + 0.25(1 - z_1) = x - 0.25z_1$ on the left. At $x = 0.5, z_1 = 1$ it gives $h_1 \leq 0.25$; the first row gives $h_1 \geq 0.25$, fixing $h_1 = 0.25$.

Conditions for MILP representability

A graph of fixed affine layers, ReLUs, and affine scaling can be expressed by an MILP. The objective and all remaining constraints must also be linear for the entire problem to be an MILP.

Products Qy with both flow and concentration variable, nonlinear kinetics, or sigmoid/tanh operations fall outside this formulation.

The Week 4 projection can also be appended as affine equalities when its coefficient matrices are fixed and its input dependence is affine.

MILP and forward-pass consistency at fixed inputs

Fix one residence time, remove product specifications, and solve the network equations. Compare every h and y with the forward pass.

41 checked inputs: domain endpoints, all breakpoints, both sides of interior breakpoints, and random inputs. Maximum difference: 1.25e-08.

This tests the formulation implementation. Process prediction error requires comparison with separate reference values.

Once the input is fixed, the original network output is determined. A different solver output points to weight ordering, scaling, signs, or bounds. Product specifications are separate from evaluating the network at that point, so remove them during this check.

CSTR design optimization results

Quantity	This run
Minimum τ [min]	5.353846
Minimum V [L]	5.353846
Predicted [A, B, C] [mol/L]	[0.584615, 0.400000, 0.215385]
Binary [z_1, z_2, z_3]	[1, 0, 0]
Termination	optimal, status = 0

The B lower bound is active, while the C upper bound has slack. Independent segmentwise linear calculations give the same residence time.

B slack is $0.400000-0.40=0$; C slack is $0.25-0.215385=0.034615$ mol/L. A small decrease in τ makes B insufficient first. This active constraint prevents a further volume reduction.

Interpolation fraction at the product lower bound

Across 4–6 min, B increases from 0.380952381 to 0.409090909. Divide the required increase by the full segment increase.

Calculation	Value
Required B increase	$0.40-0.380952381=0.019047619$
Full segment increase	$0.409090909-0.380952381=0.028138528$
Segment fraction α	$0.019047619/0.028138528\approx0.676923077$
Residence time	$4+2\alpha\approx5.353846154$ min
C at the same α	$0.152380952+\alpha(0.245454545-0.152380952)\approx0.215384615$

B is below 0.40 on all earlier segments. At this first crossing, $C\leq0.25$ and nonnegativity also hold, giving the minimum volume. Comparing interpolated B with C at a different τ would not check specifications for the same design.

Formulation equivalence and process prediction error

The MILP exactly represents the frozen interpolant. This property does not remove its process approximation error.

At the selected $\tau = 5.353846$ min, the reference concentrations are [0.579495, 0.404137, 0.216369] mol/L, which differ from the predictor.

The solution is optimal for the specified surrogate problem. A process design decision also requires checking reference-process specifications and comparing objective values.

LP relaxation and objective lower bounds

Relaxing z to [0, 1] gives a lower bound on the MILP optimum for minimization. Intermediate values and outputs may nevertheless leave the network graph.

With $L = -1$, $U = 2$, $a = 0$, the relaxed point $z = 1/3$, $h = 2/3$ is feasible, although the ReLU output is 0.

The LP and MILP objective values coincide in this CSTR example. Equality of objectives in one instance does not establish general graph equivalence.

Enlarging a minimization feasible set retains the original optimum, so the new optimal value cannot be larger. This makes the LP value a lower bound. Its outputs may leave the network graph, so an LP solution is not automatically an acceptable design.

Run the lab and prepare the submission

```
python -m pip install -r requirements.txt
python week05_lab.py --output-dir my-results
```

1. Hand checks: one neuron’s bounds and matrix signs
2. Checks in `results.json`: forward error and termination status
3. Translate one row of `milp_matrix.csv` back into an equation.
4. Change the B lower bound from 0.40 to 0.41 and explain whether feasibility remains.

reading companion · lab bundle

Sources and assigned reading

Boyd & Vandenberghe, Convex Optimization, §4.1: variables, objectives, and feasible sets.

Anderson et al., Strong mixed-integer programming formulations for trained neural networks, §2.1: the ReLU Big-M formulation.

SciPy milp documentation: bounds, integrality, LinearConstraint, and return status.